

CHAPTER 5

The d- and f-Block Elements

Transition Elements (d-Block)

1. Which of the following pairs of ions will have same spin only magnetic moment values within the pair? (2024 Re)

A. Zn^{2+} , Ti^{2+} B. Cr^{2+} , Fe^{2+}
C. Ti^{3+} , Cu^{2+} D. V^{2+} , Cu^{+}

Choose the correct answer from the options given below:

a. C and D only b. A and D only
c. A and B only d. B and C only

2. The E° value for the $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is more positive than that of $\text{Cr}^{3+}/\text{Cr}^{2+}$ or $\text{Fe}^{3+}/\text{Fe}^{2+}$ due to change of (2024)

a. d^4 to d^5 configuration b. d^3 to d^5 configuration
c. d^5 to d^4 configuration d. d^5 to d^2 configuration

3. 'Spin only' magnetic moment is same for which of the following ions? (2024)

A. Ti^{3+} B. Cr^{2+} C. Mn^{2+} D. Fe^{2+} E. Sc^{3+}

Choose the most appropriate answer from the options given below:

a. B and C only b. A and D only
c. B and D only d. A and E only

4. The stability of Cu^{2+} is more than Cu^{+} salts in aqueous solutions due to: (2023)

a. second ionisation enthalpy
b. first ionisation enthalpy
c. enthalpy of atomisation
d. hydration energy

5. Which of the following is **correctly** matched? (2023-Manipur)

a. Basic oxides $\Rightarrow \text{In}_2\text{O}_3$, K_2O , SnO_2
b. Neutral oxides $\Rightarrow \text{CO}$, NO_2 , N_2O
c. Acidic oxides $\Rightarrow \text{Mn}_2\text{O}_7$, SO_2 , TeO_3
d. Amphoteric oxides $\Rightarrow \text{BeO}$, Ga_2O_3 , GeO

6. Given below are two statements : (2022 Re)

Statement I: Cr^{2+} is oxidising and Mn^{3+} is reducing in nature.

Statement II: Sc^{3+} compounds are repelled by the applied magnetic field.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

a. Statement I is incorrect but Statement II is correct
b. Both Statement I and Statement II are correct
c. Both Statement I and Statement II are incorrect
d. Statement I is correct but Statement II is incorrect

7. Zr ($Z = 40$) and Hf ($Z = 72$) have similar atomic and ionic radii because of: (2021)

a. Diagonal relationship
b. Lanthanoid contraction
c. Having similar chemical properties
d. Belonging to same group

8. The calculated spin only magnetic moment of Cr^{2+} ion is (2020)

a. 4.90 BM b. 5.92 BM c. 2.84 BM d. 3.87 BM

9. Match the following aspects with the respective metal. (2020-Covid)

Aspects	Metal
A. The metal which reveals a maximum number of oxidation states	i. Scandium
B. The metal although placed in 3d block is considered not as a transition element	ii. Copper
C. The metal which does not exhibit variable oxidation states	iii. Manganese
D. The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	iv. Zinc

Select the correct option:

a. A-iii B-iv C-i D-ii
b. A-iii B-i C-iv D-ii
c. A-ii B-iv C-i D-iii
d. A-i B-iv C-ii D-iii

10. Match the metal ions given in Column-I with the spin magnetic moments of the ions given in Column-II and assign the correct code: (2018)

Column-I	Column-II
A. Co^{3+}	i. $\sqrt{8}$ B.M
B. Cr^{3+}	ii. $\sqrt{35}$ B.M
C. Fe^{3+}	iii. $\sqrt{3}$ B.M
D. Ni^{2+}	iv. $\sqrt{24}$ B.M
	v. $\sqrt{15}$ B.M

a. A-iv B-v C-ii D-i
b. A-i B-ii C-iii D-iv
c. A-iii B-v C-i D-ii
d. A-iv B-i C-ii D-iii

11. HgCl_2 and I_2 both when dissolved in water containing I^- ions the pair of species formed is: (2017-Delhi)
 a. HgI_2, I^- b. $\text{HgI}_2, \text{I}_3^-$ c. HgI_2, I^- d. $\text{HgI}_4^{2-}, \text{I}_3^-$
12. Which of the following processes does not involve oxidation of iron? (2015)
 a. Decolourisation of blue CuSO_4 solution by iron
 b. Formation of $\text{Fe}(\text{CO})_5$ from Fe
 c. Liberation of H_2 from steam by iron at high temperature
 d. Rusting of iron sheets
13. Magnetic moment 2.84 B.M. is given by: (Atomic numbers, Ni = 28, Ti = 22, Cr = 24, Co = 27) (2015)
 a. Ti^{3+} b. Cr^{2+} c. Co^{2+} d. Ni^{2+}
14. Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii? (Numbers in the parenthesis are atomic numbers) (2015)
 a. Zr (40) and Nb (41) b. Zr (40) and Hf (72)
 c. Zr (40) and Ta (73) d. Ti (22) and Zr (40)
15. Magnetic moment 2.83 BM is given by which of the following ions? (2014)
 (Atomic Number Ti = 22, Cr = 24, Mn = 25, Ni = 28)
 a. Ni^{2+} b. Cr^{3+} c. Mn^{2+} d. Ti^{3+}
16. Which of the following statements about the interstitial compounds is incorrect? (2013)
 a. They retain metallic conductivity
 b. They are chemically reactive
 c. They are much harder than the pure metal
 d. They have higher melting points than the pure metal
- c. The oxidation states of chromium in CrO_4^{2-} and $\text{Cr}_2\text{O}_7^{2-}$ are not the same.
 d. $\text{Cr}^{2+}(\text{d}^4)$ is a stronger reducing agent than $\text{Fe}^{2+}(\text{d}^6)$ in water.
20. The manganate and permanganate ions are tetrahedral, due to: (2019)
 a. The p-bonding involves overlap of p-orbitals of oxygen with d-orbitals of manganese
 b. There is no p-bonding
 c. The p-bonding involves overlap of p-orbitals of oxygen with p-orbitals of manganese
 d. The p-bonding involves overlap of d-orbitals of oxygen with d-orbitals of manganese
21. Which one of the following ions exhibits d-d transition and paramagnetism as well? (2018)
 a. CrO_4^{2-} b. $\text{Cr}_2\text{O}_7^{2-}$ c. MnO_4^{2-} d. MnO_4^-
22. Name the gas that can readily decolourise acidified KMnO_4 solution: (2017-Delhi)
 a. P_2O_5 b. CO_2 c. SO_2 d. NO_2
23. Which one of the following statements is correct when SO_2 is passed through acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution? (2016-I)
 a. SO_2 is reduced b. Green $\text{Cr}_2(\text{SO}_4)_3$ is formed
 c. The solution turns blue d. The solution is decolourised
24. Assuming complete ionisation, same moles of which of the following compounds will require the least amount of acidified KMnO_4 for complete oxidation? (2015 Re)
 a. $\text{Fe}(\text{NO}_2)_2$ b. FeSO_4 c. FeSO_3 d. FeC_2O_4

Compounds of Transition Elements

17. Which of the following statements are INCORRECT? (2023)
 A. All the transition metals except scandium form MO oxides which are ionic.
 B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7 .
 C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5 .
 D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.
 E. CrO is basic but Cr_2O_3 is amphoteric.
 Choose the correct answer from the options given below:
 a. B and C only b. A and E only
 c. B and D only d. C and D only
18. In the neutral or faintly alkaline medium, KMnO_4 oxidises iodide into iodate. The change in oxidation state of manganese in this reaction is from (2022)
 a. +6 to +5 b. +7 to +4 c. +6 to +4 d. +7 to +3
19. Identify the incorrect statement. (2020)
 a. The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.
 b. Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.

Inner Transition Elements (Lanthanoids and Actinoids)

25. Which of the following set of ions act as oxidising agents? (2024 Re)
 a. Ce^{4+} and Tb^{4+} b. La^{3+} and Lu^{3+}
 c. Eu^{2+} and Yb^{2+} d. Eu^{2+} and Tb^{4+}
26. The UV-visible absorption bands in the spectra of lanthanoid ions are 'X', probably because of the excitation of electrons involving 'Y'. The 'X' and 'Y', respectively, are: (2024 Re)
 a. Broad and f orbitals
 b. Narrow and f orbitals
 c. Broad and d and f orbitals
 d. Narrow and d and f orbitals
27. Match List-I with List-II: (2024 Re)

	List-I (Block/group in periodic table)		List-II (Element)
A.	Lanthanoid	I.	Ce
B.	d-block element	II.	As
C.	p-block element	III.	Cs
D.	s-block element	IV.	Mn

- Choose the correct answer from the options given below:
 a. A-I, B-II, C-IV, D-III b. A-I, B-IV, C-III, D-II
 c. A-I, B-IV, C-II, D-III d. A-IV, B-I, C-II, D-III

28. The pair of lanthanoid ions which are diamagnetic is (2024)
 a. Gd^{3+} and Eu^{3+} b. Pm^{3+} and Sm^{3+}
 c. Ce^{4+} and Yb^{2+} d. Ce^{3+} and Eu^{2+}

29. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Ionisation enthalpies of early actinoids are lower than for early lanthanoids.

Reason (R): Electrons are entering 5f orbitals in actinoids which experience greater shielding from nuclear charge.

In the light of the above statements, choose the **correct** answer from the options given below: (2023-Manipur)

- a. (A) is true but (R) is false.
 b. (A) is false but (R) is true.
 c. Both (A) and (R) are true and (R) is the correct explanation of (A).
 d. Both (A) and (R) are true but (R) is **not** the correct explanation of (A).
30. Decrease in size from left to right in actinoid series is greater and gradual than that in lanthanoid series due to: (2022 Re)
 a. 5f orbitals have greater shielding effect
 b. 4f orbitals are penultimate
 c. 4f orbitals have greater shielding effect
 d. 5f orbitals have poor shielding effect
31. Gadolinium has a **low** value of third ionisation enthalpy because of (2022)
 a. high basic character
 b. small size
 c. high exchange enthalpy
 d. high electronegativity
32. The incorrect statement among the following is: (2021)
 a. Most of the trivalent Lanthanoid ions are colorless in the solid state
 b. Lanthanoids are good conductors of heat and electricity
 c. Actinoids are highly reactive metals, especially when finely divided
 d. Actinoid contraction is greater for element to element than Lanthanoid contraction
33. Identify the incorrect statement from the following: (2020-Covid)
 a. Lanthanoids reveal only +3 oxidation state.
 b. The lanthanoid ions other than the f^0 type and the f^4 type are all paramagnetic.
 c. The overall decreases in atomic and ionic radii from lanthanum to lutetium is called lanthanoid contraction.
 d. Zirconium and Hafnium have identical radii of 160 pm and 159 pm, respectively as a consequence of lanthanoid contraction.

34. Match the element in Column-I with that in Column-II. (2020-Covid)

Column-I		Column-II	
A.	Copper	i.	Non-metal
B.	Fluorine	ii.	Transition metal
C.	Silicon	iii.	Lanthanoid
D.	Cerium	iv.	Metalloid

Select the correct option:

- a. A-i B-ii C-iii D-iv
 b. A-ii B-iv C-i D-iii
 c. A-ii B-i C-iv D-iii
 d. A-iv B-iii C-i D-ii
35. The reason for greater range of oxidation states in actinoids is attributed to: (2017-Delhi)
 a. 4f and 5d levels being close in energies
 b. The radioactive nature of actinoids
 c. Actinoid contraction
 d. 5f, 6d and 7s levels having comparable energies
36. Which one of the following statements related to lanthanons is incorrect? (2016-II)
 a. All the lanthanons are much more reactive than aluminium
 b. $Ce(+4)$ solutions are widely used as oxidizing agent in volumetric analysis
 c. Europium shows +2 oxidation state.
 d. The basicity decreases as the ionic radius decreases from Pr to Lu.
37. The electronic configurations of Eu (Atomic Number 63) Gd (Atomic Number 64) and Tb (Atomic Number 65) are: (2016-I)
 a. $[Xe]4f^7 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^9 6s^2$
 b. $[Xe]4f^7 6s^2$, $[Xe]4f^8 6s^2$ and $[Xe]4f^8 5d^1 6s^2$
 c. $[Xe]4f^6 5d^1 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^9 5d^1 6s^2$
 d. $[Xe]4f^6 5d^1 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^8 5d^1 6s^2$
38. Gadolinium belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium? (2015 Re)
 a. $[Xe]4f^6 5d^2 6s^2$ b. $[Xe]4f^8 6d^2$
 c. $[Xe]4f^9 5s^1$ d. $[Xe]4f^7 5d^1 6s^2$
39. Reason of lanthanoid contraction is: (2014)
 a. Increasing nuclear charge
 b. Decreasing nuclear charge
 c. Decreasing screening effect
 d. Negligible screening effect of 'f' orbitals
40. Which of the following lanthanoid ions is diamagnetic? (Atomic Number Ce = 58, Sm = 62, Eu = 63, Yb = 70) (2013)
 a. Ce^{2+} b. Sm^{2+} c. Eu^{2+} d. Yb^{2+}

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (c) | 3. (c) | 4. (d) | 5. (c) | 6. (a) | 7. (b) | 8. (a) | 9. (a) | 10. (a) |
| 11. (d) | 12. (b) | 13. (d) | 14. (b) | 15. (a) | 16. (b) | 17. (d) | 18. (b) | 19. (c) | 20. (a) |
| 21. (c) | 22. (c) | 23. (b) | 24. (b) | 25. (a) | 26. (b) | 27. (c) | 28. (c) | 29. (c) | 30. (d) |
| 31. (c) | 32. (a) | 33. (a) | 34. (c) | 35. (d) | 36. (a) | 37. (a) | 38. (d) | 39. (d) | 40. (d) |

